

**The Politics of Reachability:
Institutional Landscapes and the Distribution of Agency**

PIATRA . INSTITUTE

August 2026

Abstract

Political theory measures freedom by non-interference, by resources, or by substantive opportunity. A prior fact escapes all three: institutions construct the paths through which intentions become consequential action, and those paths, their costs, their risks, and their gatekeepers are objects of power in their own right. This paper develops an account of freedom as reachability, the set of consequential states an agent can actually attain from its position through existing institutional pathways within its budget of money, time, risk, and permission, and it builds the account's core objects in a simulation where they can be computed. In a landscape of 6 pathways permitted identically to every agent, with a cheap consumption state reachable by all so that option-counting sees no problem anywhere, the reachable set runs from 6 pathways for the top budget decile to 0 for the bottom five, and the pathway modeled on open foundational infrastructure, permitted to everyone, is reachable by 3 deciles of 10: the missing valley is an object, and it is missing for almost everyone. Fed back on itself, the landscape narrows endogenously: returns from the one appropriable pathway deepen it and silt its alternatives, and over 60 generations the entropy of chosen pathways falls from 1.69 bits to 0.21 with no change in law, while the deepening canal genuinely admits more deciles than it began with. Capping what returns may do to rival pathways preserves the reachability of alternatives and leaves the canal's share concentrated anyway; the cap protects the exit, never the market share. An incumbent spending returns on ridge-raising and office-capture shrinks the reachable sets of 5 deciles while every consumer price in the model stays fixed, which is what an option-space conception of competition sees and a price conception cannot. A shared substrate raises reach wherever it exists, and both substrates canalize, routing over 0.9 of attempts through themselves; the open one leaves the larger field, 49 pathway-deciles against 45, and survives its steward's failure with 1.0 of its reach where the enclosed platform retains 0.56; the enclosed substrate wins in one regime, the world where nobody funds maintenance, because the gatekeeper at least pays the maintainers. A floor under failure moves the capacity to attempt anything consequential from 5 deciles to 7; the same floor in a flat world moves it from 4 to 5, because floors need valleys. The politics that follows has two tasks that must be performed together: bound any actor's unilateral power over the landscape, and build the plural, generative, forkable pathways through which ordinary reach expands. The opposite of domination is a landscape nobody can privately freeze and everyone has some effective capacity to reshape.

The missing valley

An engineer wants to build a new foundational network architecture: open, interoperable, designed to resist enclosure, likely to generate enormous social value and no monopoly rent. No law forbids the project. No censor waits at the end of it. And in most of the developed world the project has almost nowhere to live. There is no standard funding form for it, no organizational template, no maintenance financing, no procurement route, and no way for its builders to eat while building. The engineer is free to attempt it in every sense the classical theories of freedom can express, and the attempt is institutionally stillborn. Meanwhile one pathway stands open, deep, and well-lit: prototype, accelerator, venture round, proprietary firm, scale, exit. The project that enters it will

not be the project the engineer meant to build.

The paper is about the difference between those two facts. A path can be permitted and absent. A person can hold rights, and even resources, and stand in a landscape where nothing consequential is reachable from where they are. And the structure that decides which paths exist, what they cost, and who waits at their gates is built, maintained, and rebuilt by identifiable institutions, which means it can be held, fought over, and owned. Freedom, on the account developed here, is reachability: the set of consequential states an agent can actually attain, from its position, through existing pathways, within its budget of money, time, risk, and permission. The account owes three things: a concept that separates reachability from its predecessors, a theory of the power that operates on landscapes rather than on persons, and a criterion for telling good landscapes from bad ones. It delivers them with a model in which every one of its objects is computed.

Freedom as reachability

Negative liberty asks whether anyone interferes (Berlin, 1969). Nobody interferes with the engineer. Resource accounts ask what a person holds, and the same holdings buy different lives in different institutional environments, which is why the question was always incomplete. The capability approach comes closest: Sen relocated freedom from holdings to substantive opportunity, what people are actually able to do and be (Sen, 1999). Reachability is a capability concept made dynamic and institutional. Three additions do the work.

Opportunity is path-shaped. Between an agent and any consequential state stands a sequence of transitions, each with a cost, a risk, a duration, and sometimes a gatekeeper, and the feasibility of the destination is a property of the whole sequence. A state can be unreachable although every individual step is permitted, because the steps do not exist as a fundable, survivable series. Gibson's affordances name the ancestor notion at the scale of an organism and its surfaces: what an environment offers an agent, relative to what that agent is (Gibson, 1979). Reachability is affordance at the scale of institutions, where the surfaces are legal forms, funding instruments, and standards, and the offering runs over years.

Consequential states are mostly reached in company. Nobody individually builds a network, a university, or a scientific discipline; the reachable set of an isolated person contains almost nothing that matters here. Access to organization is therefore a component of freedom rather than an external convenience, and a landscape that starves every organizational form except one has restricted freedom even if each person's individual options are untouched.

And a path that runs through discretion differs in kind from a path that runs on right. A pathway passable only when an office says yes places the traveler under the office's arbitrary power, which is Pettit's definition of domination, and it remains domination when the office is benevolent and the yes always comes (Pettit, 1999). The model below keeps the two kinds of reach in separate columns for this reason.

Institutional landscapes

Waddington drew development as a ball rolling through a landscape of valleys, whose shape, produced by the interactions underneath, makes some trajectories easy, some hard, and some effectively impossible; canalization is his name for the deepening of a valley until development returns to it despite perturbation (Waddington, 1942; Waddington, 1957). The figure transfers to institutions as structure rather than as biology. The landscape is the transition system produced by property and corporate law, finance, intellectual-property rules, procurement, standards, credentialing, social insurance, and infrastructure; a valley is a viable organizational trajectory; a ridge is a barrier of cost, law, or legibility; an attractor is a stable form that projects converge upon.

One disanalogy carries the politics and must be stated at the start. An embryo does not lobby for its landscape. Social landscapes are reflexive: the agents developing inside them can discover their shape, contest it, and rebuild it, and the most successful travelers acquire the means to reshape the terrain behind them. Institutions do not determine conduct mechanically; they differentially weight the cost, risk, intelligibility, and expected viability of strategies, which is why the landscape figure needs a mechanism rather than a mood. Arthur supplied it: under increasing returns, early advantage compounds, adoption begets adoption, and a technology or an institutional form can lock in against alternatives that would have served at least as well (Arthur, 1989). Canalization is path dependence read topographically.

Law sits underneath the whole surface. Pistor's account of capital is an account of landscape construction: assets become capital by being coded in legal modules, priority, durability, universality, convertibility, and the coding profession decides which projects acquire the armor (Pistor, 2019). Polanyi's old point stands behind it: the market order is itself an institutional construction, laid down and maintained, and there was never a state of nature for regulation to disturb (Polanyi, 1944). There is no landscape-free baseline. The question is never whether to shape the terrain; it is who shapes it and toward what.

The corporate canal

The Internet is the modern demonstration that the dominant pathway is one design among several, because the foundational layer of the present economy was built almost entirely outside it. Packet internetworking was published as an open protocol by researchers on public contracts (Cerf and Kahn, 1974). The network grew through defense agencies, national science foundations, universities, and an open standards process in which anyone could submit a draft; the participants' own history is a history of public funding and institutional plurality (Leiner et al., 2009). The Web was written at a physics laboratory, and on 30 April 1993 CERN put the software into the public domain (European Organization for Nuclear Research, 1993). The architecture that resulted did not convert the value it created into a claim on everything downstream. The protocols became substrate, and Google, Wikipedia, a million firms, and several billion people built above them. This is why the inventors of TCP/IP and the Web are not the richest people in history, and why that fact indicts nothing about them and describes the landscape they worked in.

None of this was purity. Military money, contractors, telephone monopolies, and later platform capital ran through every stage, and the point survives the mess: a plural landscape with an open foundational layer produced infrastructure that no venture pathway would have built. Consider what the venture form selects for. Its return structure needs appropriability, rapid scale, defensibility, concentrated ownership, and an exit inside the fund's horizon. A project can fail every one of those tests and be exactly what a civilization needs: open by design, slow to mature, rent-free, generative of value it cannot capture. The venture verdict on such a project, a terrible company, is fully compatible with the verdict excellent infrastructure, and a landscape whose only deep valley is the venture canal will systematically not build its own foundations. The claim is comparative, and the model makes it without any villain: nobody in what follows breaks a law, and the narrowing happens anyway.

The model

The landscape is built literally. Six pathways run from a common start to consequential terminals: a venture canal, cheap to enter and appropriable at the end; a discretionary grant path; a commons path; a cooperative path; a public-agency path behind an office; and the opening case, open foundational infrastructure, long, subsidized by nothing, owned by nobody at the end. Beside them sits a consumption state, cheap and inconsequential, so that a theory counting options sees no problem anywhere. Each pathway is a sequence of stage costs with a failure risk; two carry permission gates. Agents differ in one respect only, budget, arranged in 10 deciles from 1.0 to 40.0, and an uninsured agent must hold a ruin reserve of 2.0 against failure before it can attempt anything. Every pathway is legally open to every agent in every experiment below; permission, in the legal sense, never varies. The model is a demonstration of structure, its numbers are facts about its geometry, and the paper claims nothing empirical through them.

Four experiments sit on the landscape. The first computes reachable sets by decile. The second runs 60 generations in which agents choose affordable pathways by expected value and the returns of the appropriable terminal work the landscape: they deepen the venture canal and, unless capped, raise the costs of its rivals. The third hands an incumbent a budget of landscape operations and holds every price fixed. The fourth builds three worlds, one with no shared substrate, one on an open protocol, one on an enclosed platform, and breaks their stewards.

What the landscape returns

Permission is flat and reachability is a staircase. The top decile reaches all 6 pathways, 4 outright and 2 through an office's discretion. The bottom 5 deciles reach nothing consequential at all, while the consumption state stays reachable by everyone, which is the configuration the classical theories cannot see: no interference, universal permission, options everywhere, and half the population positioned where no consequential state is attainable. The open-infrastructure pathway, permitted to all, is reachable by deciles 8 through 10. The engineer of the opening section is, in this landscape, a member of decile 7: able to reach the venture canal, the commons, and the cooperative form, and standing before the one valley that fits the project, which exists in law and not in fact.

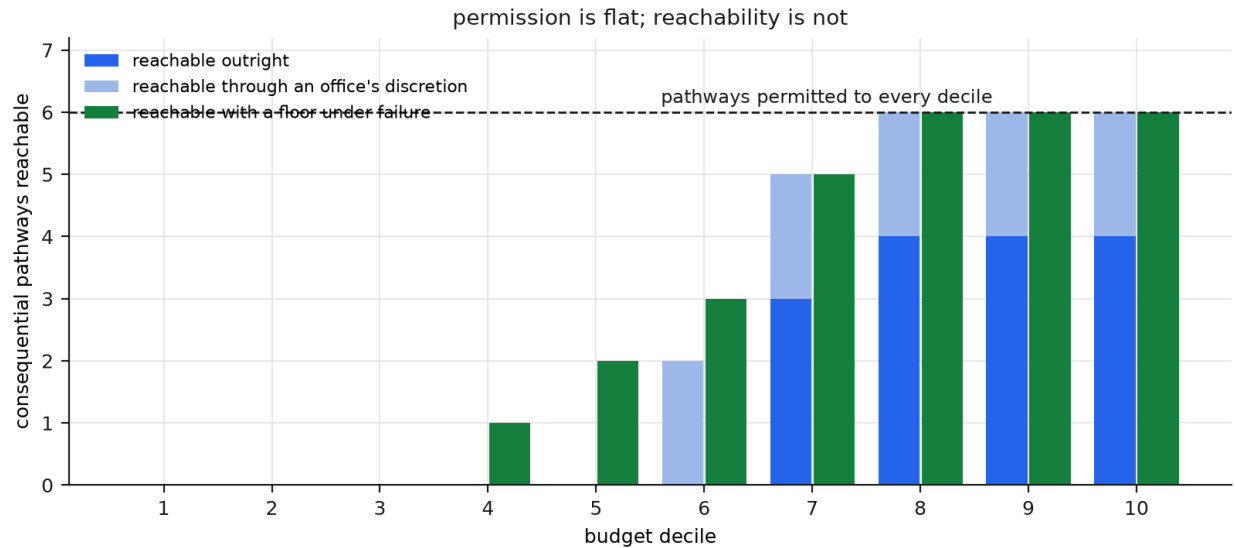


Figure 1: Reachability by budget decile in a landscape whose 6 pathways are permitted to every decile. The bottom half reaches nothing consequential; the open-infrastructure path is reachable by 3 deciles; a floor under failure moves the frontier down the distribution.

The second experiment lets the landscape move. Returns from the venture terminal deepen the canal and silt the alternatives, and over 60 generations the entropy of chosen pathways falls from 1.69 bits to 0.21, the canal's share of attempts rising from 0.59 to 0.97, with no statute altered and no gate closed. Two honesty clauses belong to this result. The canal is genuinely enabling while it narrows: its deepening admits a fifth decile that could never afford it at the start, which is why the narrowing is experienced from inside as opportunity. And no malice is required for the concentration: it is arithmetic, Arthur's compounding run through a terrain.

The cap experiment separates two things the politics of this subject habitually confuses. Bounding what returns may do to rival pathways, leaving self-improvement unbounded, does not stop the canal from winning the traffic; its share concentrates above 0.9 anyway, because it keeps getting better. What the cap preserves is the reachability of the alternatives: 18 pathway-deciles outside the canal survive at generation 60 under the cap, against 15 without it, and the uncapped landscape is still silting when the run ends. On the paper's own account of freedom this is the entire point. Freedom was never the market share of the alternatives; it is their reachability when someone wants one. The cap protects the exit, never the share.

Landscape power

The third experiment gives the incumbent at the top of the canal a budget of operations: ridges raised over the commons, the cooperative, and the infrastructure paths, and the grant office absorbed, its gate now opening on the incumbent's account. Every consumer price in the model is identical before and after. The reachable sets of 5 deciles shrink, the harm landing heaviest on the middle of the distribution, deciles 6 through 8, whose members lose most of what they had

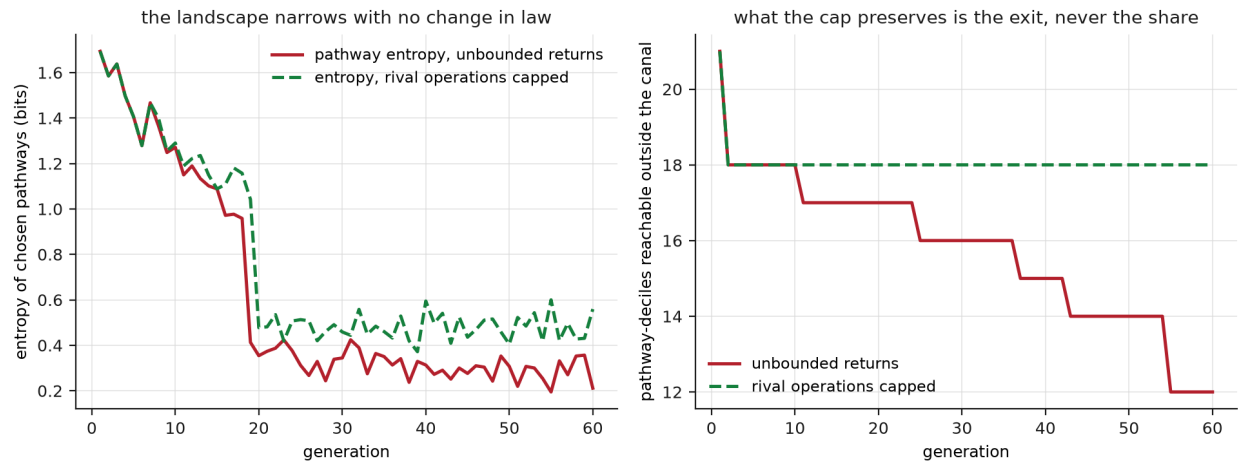


Figure 2: Left: entropy of chosen pathways across 60 generations, with returns unbounded and with rival operations capped; no law changes in either run. Right: the pathway-deciles reachable outside the canal, the quantity the cap actually protects.

recently gained, while the top decile loses one pathway and keeps five. An antitrust doctrine that measures consumer prices reads this world as harmless. An option-space doctrine reads it directly: the damage of concentrated private power is counted in deleted futures, in the adjacent possible destroyed (Kauffman, 2000), and the price column is where the damage does not appear.

This is the level of power the tradition has been circling. Strange called structural power the capacity to shape the frameworks within which others must operate (Strange, 1988); Deleuze described control as modulation, power exercised on the medium rather than the act (Deleuze, 1992). Landscape power is the dynamic, topological form of the same object: the capacity to alter the transition structure through which other agents pursue consequential action, to create and delete pathways, reprice them, gate them, and freeze them. It is second-order agency, power over the conditions of everyone else's first-order agency, and it is the correct register for the question about extreme wealth. Winters and Page showed that oligarchy is a fact about material power concentration and coexists comfortably with elections (Winters and Page, 2009; Winters, 2011); Anderson showed that liberal societies police arbitrary authority everywhere except inside the institutions they call private (Anderson, 2017); Rawls conceded that the political liberties lose their fair value under sufficient economic inequality (Rawls, 1993); Robeyns built the limitarian case that extreme wealth is objectionable on democratic grounds independent of desert (Robeyns, 2019).

The reachability frame sharpens all four into one question. The objection to a billionaire is misdrawn as a number on a balance sheet; wealth at that scale is a general-purpose converter into landscape operations, into standards, infrastructure, media, lobbying, and law, and the constitutional question is how large a change in other people's reachable sets one person may produce unilaterally. Societies already answer that question for every officer of the formal constitution: presidents, generals, judges, and reactor operators act inside authorization machinery precisely because their reach is consequential. The same societies leave cross-domain private reach almost

unmachined, because it arrives described as property. The principle the model's cap implements is bounded landscape power: improvement of one's own valley unbounded, operations on other people's pathways constrained, and no actor positioned to freeze the terrain. Nothing in the principle depends on the operator's character. Constitutions are what a polity builds when it stops relying on the virtue of whoever currently holds the handle.

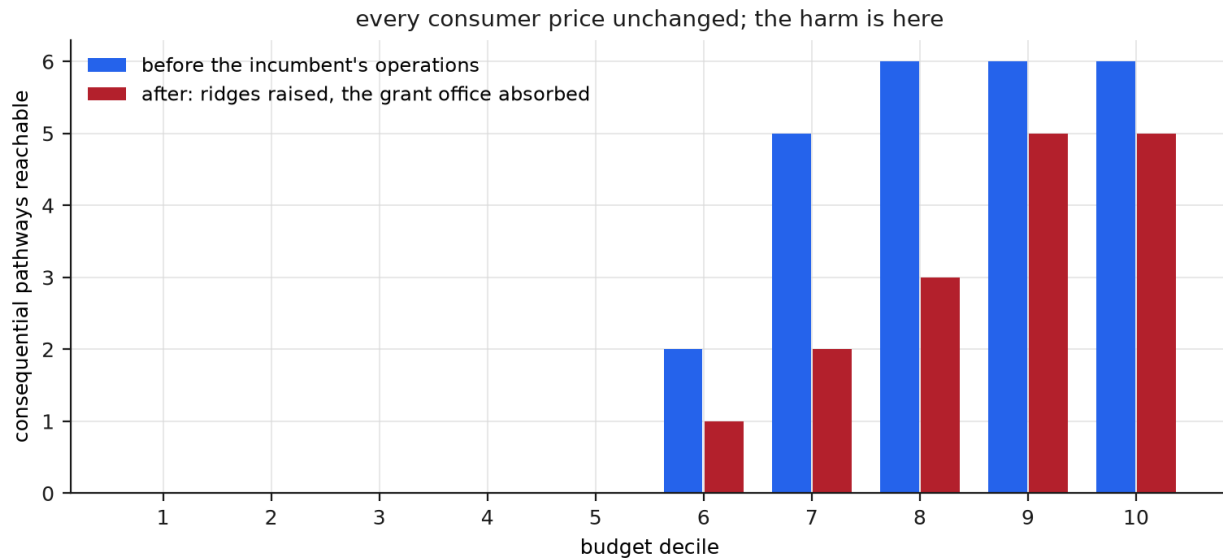


Figure 3: The incumbent's operations: reachable pathways by decile, before and after, with every consumer price unchanged. The harm is real, large, unevenly distributed, and invisible to any doctrine that audits prices.

Substrates

The fourth experiment tests the positive side: what shared foundations do. Against the flat world, where nothing is shared and every pathway carries the full cost of its own infrastructure, a substrate world adds 4 downstream pathways spawned by the foundation, open in one variant, tolled and gated by the platform in the other. Reach in the flat world is 19 pathway-deciles. The open substrate raises it to 49, the enclosed one to 45. And both substrates canalize: over 0.9 of attempts route through the substrate-spawned paths in either world, 0.97 in the open one. TCP/IP is also a canal. The claim that all canalization is domination dies here, and Zittrain's criterion replaces it: the generative substrate is the one that lets audiences its designers never anticipated build things its designers never imagined (Zittrain, 2006). The distinction between the two substrates is a single property, who may build on the foundation without asking, and the crisis experiment prices it. When the substrate's steward fails, the open world retains 1.0 of its reach, because the downstream paths are forkable and survive their origin, while the enclosed world retains 0.56, because the gate dies with its keeper.

One regime rescues the platform, and honesty requires it reported with the same prominence. Open substrates need maintainers, and maintenance is the commons problem in its least romantic

form. In the model's low-funding regimes the open substrate silts, and at zero public maintenance the enclosed platform strictly beats it, 45 pathway-deciles to 41, with the two tying across the middle of the funding range; the open substrate wins only where maintenance is actually paid. The gatekeeper, whatever else it takes, pays the maintainers out of the tolls. A polity that wants open foundations and will not fund stewardship has chosen the platform without admitting it. Ostrom is the standing demonstration that stewardship without an owner is an institutional craft with known working designs rather than a paradox (Ostrom, 1990; Ostrom, 2010), and Mazzucato is the record that the deepest substrates of the present economy were built by public risk-taking that the resulting private fortunes have preferred to forget (Mazzucato, 2013).

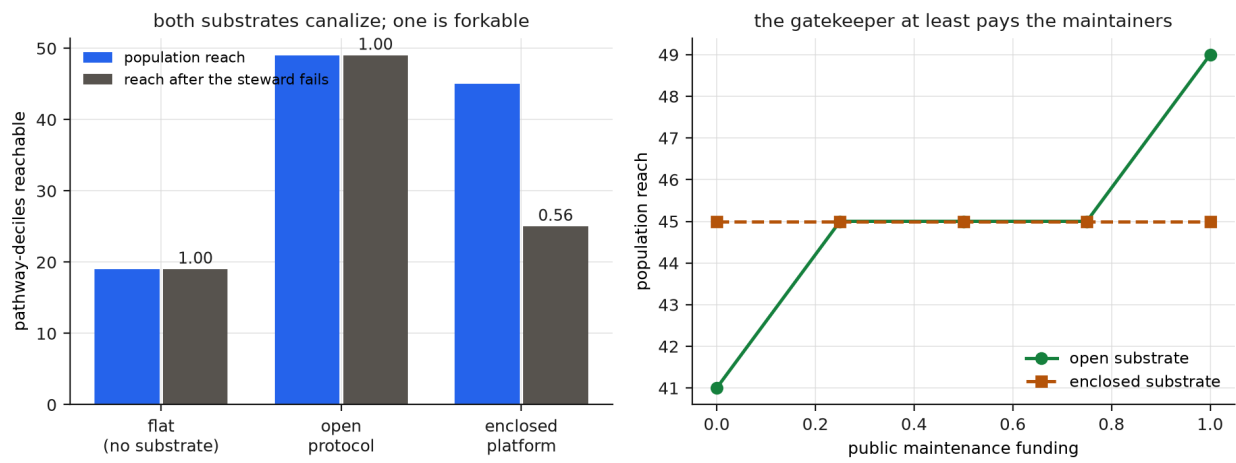


Figure 4: Left: population reach in the three worlds, and what survives the steward's failure; both substrates canalize, and one is forkable. Right: the maintenance regime, where the enclosed platform wins at zero funding and the open substrate only where stewardship is paid.

The floor

The last measured result concerns the bottom of the distribution. An uninsured agent must reserve against ruin before attempting anything, so ruin risk functions as a surcharge on every consequential pathway, and the surcharge prices out the poor from experiments the rich run casually. Installing a floor, insurance against the failure of an attempt, moves the number of deciles able to attempt any consequential pathway from 5 to 7. Social provision, read through this result, is risk-bearing infrastructure: not a transfer that compensates after the market has distributed, but a change in which trajectories exist for whom. A person two missed paychecks from homelessness cannot spend 2 years building an open protocol, and no amount of formal permission alters that arithmetic.

The companion result disciplines the enthusiasm. The same floor installed in the flat world moves the frontier from 4 deciles to 5, because there is little to attempt; insurance against failure is worth what the reachable pathways make it worth. Floors need valleys. A welfare politics that funds security without building pathways, and an innovation politics that digs pathways reachable only from the top deciles, are each holding half of one policy.

Designing for forkability

The design principles fall out of the model's own operations, and each is an existing instrument somewhere. Lower entry ridges: the floor, plus universal services, is the reachability version of social policy. Create ownerless valleys: funding forms that do not manufacture an owner, grants, fellowships, prizes, mission agencies, are the missing valley dug on purpose, and their present scale is a rounding error beside the canal. Keep the foundational layer open and let competition happen above it: the strongest single rule the genealogy supports, and the one the European interoperability obligations have begun, in a small way, to legislate (European Union, 2022). Fund stewardship as infrastructure, since the maintenance regime is where open foundations are won or lost. Bound landscape power: structural limits on cross-domain control and on operations against rival pathways, which is antitrust reconceived as landscape maintenance, keeping valleys from becoming canyons. And keep the landscape revisable: Unger's institutional experimentalism, the refusal to treat any arrangement as the necessary form of an ideal (Unger, 1987), given procedure by Sabel and Zeitlin's recursive framework of provisional goals, decentralized variation, and revision (Sabel and Zeitlin, 2008).

Hayek belongs in this section, turned once. The argument from dispersed knowledge is correct and cuts deeper than its usual conclusion: no planner knows which valleys a society will need (Hayek, 1945). A monoculture of discovery procedures is exactly what the argument forbids, and the identification of discovery with one organizational form, the investor-owned firm in a market for control, is a monoculture. The knowledge argument, taken seriously, demands plural discovery machinery: markets, commons, mission agencies, cooperatives, sciences, each canalizing differently, none owning the terrain.

Objections, and no final landscape

Every institution canalizes. Correct, and the paper's criterion never was flatness: an utterly flat landscape coordinates nothing and reaches almost nothing, as the model's flat world shows. The distinction is generative against enclosing, forkable against frozen, and both substrates in the model canalize while only one survives its steward.

More options are not more freedom. Also correct. The consumption state is reachable by everyone in every experiment and counts for nothing here; reachability is weighted toward consequential states, and the weighting is the concept, its ancestry running through Sen's refusal to value formal opportunity as such.

Standardization is efficient, and sometimes a monopoly is too. Sometimes it is, and the question the model prices is narrower: whether the coordination requires an owner with operations over rival pathways, and what the arrangement retains when its steward fails or turns. The efficiency of the gatekeeper in the unmaintained regime is real, is reported above, and is an argument for funding maintenance, and for nothing else.

Who decides which futures matter? No one finally, and that is the answer rather than an evasion. The politics defended here is procedural: bound the power that freezes, fund the floors and the

substrates, keep the forks open, and let the catalogue of desirable futures stay unwritten. A democracy on this account is a polity in which citizens do more than choose governors inside an inherited landscape; the power to reshape the landscape is itself distributed, contestable, and bounded. The opposite of domination is a terrain nobody can privately freeze and everyone can, from somewhere, begin to dig.

Reproducibility

The simulation, its four figures, and every model number quoted above regenerate from one command in the paper's repository (`simulation/run_all.py`; agent choice runs on a recorded seed, everything else is expected-value arithmetic). Fourteen invariants, among them the flatness of permission against the staircase of reach, the endogenous entropy collapse with no change of law, the cap preserving alternatives while the canal's share concentrates anyway, the zero-price shrinkage of 5 deciles' reachable sets, the canalization of both substrates, the forkable world's survival of its steward, the enclosed platform's strict win at zero maintenance, and the floor's dependence on valleys, fail the run loudly if broken.

References

- Anderson, E. (2017). *Private Government: How Employers Rule Our Lives (and Why We Don't Talk about It)*. Princeton: Princeton University Press.
- Arthur, W. B. (1989). Competing technologies, increasing returns, and lock-in by historical events. *The Economic Journal*, 99(394), 116–131.
- Berlin, I. (1969). *Four Essays on Liberty*. Oxford: Oxford University Press.
- Cerf, V. G., and Kahn, R. E. (1974). A protocol for packet network intercommunication. *IEEE Transactions on Communications*, 22(5), 637–648.
- Deleuze, G. (1992). Postscript on the societies of control. *October*, 59, 3–7.
- European Organization for Nuclear Research (1993). *Statement Concerning CERN W3 Software Release into Public Domain*. 30 April 1993. Geneva: CERN Document Server, record 1164399.
- European Union (2022). Regulation (EU) 2022/1925 on contestable and fair markets in the digital sector (Digital Markets Act). *Official Journal of the European Union*, L 265.
- Gibson, J. J. (1979). *The Ecological Approach to Visual Perception*. Boston: Houghton Mifflin.
- Hayek, F. A. (1945). The use of knowledge in society. *The American Economic Review*, 35(4), 519–530.
- Kauffman, S. A. (2000). *Investigations*. New York: Oxford University Press.
- Leiner, B. M., Cerf, V. G., Clark, D. D., Kahn, R. E., Kleinrock, L., Lynch, D. C., Postel, J., Roberts, L. G., and Wolff, S. (2009). A brief history of the internet. *ACM SIGCOMM Computer Communication Review*, 39(5), 22–31.

- Mazzucato, M. (2013). *The Entrepreneurial State: Debunking Public vs. Private Sector Myths*. London: Anthem Press.
- Ostrom, E. (1990). *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge: Cambridge University Press.
- Ostrom, E. (2010). Beyond markets and states: polycentric governance of complex economic systems. *American Economic Review*, 100(3), 641–672.
- Pettit, P. (1999). *Republicanism: A Theory of Freedom and Government*. Oxford: Oxford University Press.
- Pistor, K. (2019). *The Code of Capital: How the Law Creates Wealth and Inequality*. Princeton: Princeton University Press.
- Polanyi, K. (1944). *The Great Transformation: The Political and Economic Origins of Our Time*. New York: Farrar and Rinehart.
- Rawls, J. (1993). *Political Liberalism*. New York: Columbia University Press.
- Robeyns, I. (2019). What, if anything, is wrong with extreme wealth? *Journal of Human Development and Capabilities*, 20(3), 251–266.
- Sabel, C. F., and Zeitlin, J. (2008). Learning from difference: the new architecture of experimentalist governance in the EU. *European Law Journal*, 14(3), 271–327.
- Sen, A. (1999). *Development as Freedom*. New York: Alfred A. Knopf.
- Strange, S. (1988). *States and Markets*. London: Pinter.
- Unger, R. M. (1987). *False Necessity: Anti-Necessitarian Social Theory in the Service of Radical Democracy*. Cambridge: Cambridge University Press.
- Waddington, C. H. (1942). Canalization of development and the inheritance of acquired characters. *Nature*, 150(3811), 563–565.
- Waddington, C. H. (1957). *The Strategy of the Genes*. London: Allen and Unwin.
- Winters, J. A. (2011). *Oligarchy*. Cambridge: Cambridge University Press.
- Winters, J. A., and Page, B. I. (2009). Oligarchy in the United States? *Perspectives on Politics*, 7(4), 731–751.
- Zittrain, J. L. (2006). The generative internet. *Harvard Law Review*, 119, 1974–2040.